
AgentLoop: Goal Contracts and Deliverable-First Control for Evidence-Bound Autonomous Research

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Abstract

Autonomous coding agents can execute long research workflows, but larger agent graphs can spend more compute while making scientific claims harder to audit. We present AgentLoop, an evidence-bound control method that treats a goal as a contract over the desired outcome, allowed inputs, constraints, verification, and blocked claims. AgentLoop combines an early runnable-artifact checkpoint with immutable execution receipts, external evaluation, causal ablations, and fail-closed promotion gates. In a frozen real-model study, both stock native Codex and a two-candidate reviewer-repair portfolio produced no submission, whereas a stage-gated candidate and a single deliverable-first ablation each produced a runnable package and scored 65; the extra portfolio, review, and repair work added no score or artifact improvement. The produced numbers were not fully correct and the compute match narrowly missed its pre-registered tolerance, so AgentLoop correctly withheld a superiority claim. This study shows how artifact-first control and explicit evidence boundaries can convert autonomous research from an opaque sequence of calls into a falsifiable, recoverable scientific process.

1. From Task Agents to Research Agents

Coding agents are increasingly able to inspect repositories, call tools, edit programs, and run tests. Research requires a stricter loop: pose a question, state a falsifiable hypothesis, execute an experiment, preserve evidence, and revise the hypothesis rather than

the result. Ralphthon frames this transition as delegating not only tasks but innovation itself. Its recommended goal template specifies an outcome, evidence, constraints, allowed inputs, iteration policy, and a blocked stop condition (Team Attention, 2026).

Long-horizon execution makes this contract important. Context compaction, quota resets, provider failure, process interruption, and human steering can all occur between hypothesis and result. A system may retain a fluent narrative while losing the experiment that justifies it. Conversely, extensive orchestration can appear rigorous while consuming the budget before any executable artifact exists. We study the following question: which control mechanisms improve the survival and auditability of autonomous research deliverables, independently of model scale?

We present AgentLoop, a method rather than a fixed multi-agent topology. Its central design choice is that agent graphs are disposable hypotheses; only externally verified improvements are retained. The contribution is threefold. First, a goal contract binds desired end state, constraints, and verification before execution. Second, a deliverable-first frontier preserves the earliest runnable vertical slice before optional review. Third, a fail-closed evidence ledger prevents incomplete, contaminated, or compute-mismatched trials from becoming performance claims.

AgentLoop complements work on interleaved reasoning and acting (Yao et al., 2023), verbal self-correction (Shinn et al., 2023), and executable skill accumulation (Wang et al., 2023). Software-engineering agents show that tool interfaces can strongly affect repository-level performance (Yang et al., 2024), while SWE-bench makes executable tests the arbiter of issue resolution (Jimenez et al., 2024). Multi-agent systems such as MAGIS allocate specialized roles to repository repair (Tao et al., 2024), and Agent Laboratory assigns agents across a research pipeline (Schmidgall et al., 2025). Our question is orthogonal: when does an additional role produce evidence that justifies its cost? We treat topology as an intervention, preserve the earliest executable claim, and separate artifact survival from

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scientific correctness.

2. AgentLoop

2.1. Goal contract and intervention record

Before a run, AgentLoop freezes the task and source hashes, model and effort, tool permissions, budget, evaluator, and acceptance gates. A candidate record names the proposed mechanism, invariant, intervention, compute contract, ablation, falsifier, and forbidden task-specific localization. The baseline is current stock native goal execution, not a weaker one-shot model. A wrapper-matched ablation disables only the proposed intervention.

Formally, a goal contract is

$$G = (Y, E, C, I, S), \quad (1)$$

where Y is the required end state, E the evidence predicate, C the resource and safety constraints, I the allowed inputs and tools, and S the stop or blocked condition. For arm a , an immutable receipt ρ_a binds G to the source, executable, model, budget, artifact tree, and external verdict. Let $V(\rho_a)$ require a runnable artifact, a completed evaluator, matched frozen inputs, and an uncontaminated execution. A mechanism is eligible for promotion only if

$$V(\rho_b) \wedge V(\rho_a) \wedge C(\rho_b, \rho_a) \wedge \Delta(a, b) > 0 \wedge R(a), \quad (2)$$

where C is the preregistered compute match and R requires replication on a second task family and a sealed unseen task. A false conjunct yields a diagnostic record, not a win.

This converts “keep trying” into a scientific protocol. A run may end in a measured win, loss, or invalid comparison. Only task/source/model/compute mismatch, protected-answer exposure, unrunnable evaluation, or arm-specific infrastructure failure blocks interpretation. Unit tests, parser hardening, and provider connectivity are engineering evidence, never performance evidence.

2.2. Deliverable-first frontier

Many long runs investigate broadly before producing the file on which success depends. AgentLoop instead requests a minimal end-to-end artifact early, snapshots it, and evaluates it through a public contract. Later candidates can replace the frontier only when they preserve runnable behavior. Review is skipped when there is no artifact to review, and repair requires a concrete review file. This is an artifact-survival policy, not an answer-specific hint.

2.3. Evidence and recovery boundary

Each stage emits append-only receipts binding prompt, source, executable, model, tokens, wall time, artifact tree, and evaluator result. Hidden evaluation is isolated from generation and candidate selection. Quota or provider substitutions may help development, but they cannot be silently reclassified as a matched trial. After interruption, the controller resumes from the last verified artifact and the original goal contract. Promotion requires repeated tasks and a sealed unseen confirmation; a single successful cell remains diagnostic.

The controller therefore follows a small state machine: freeze the contract; execute an arm until a runnable checkpoint or budget stop; evaluate a content-addressed snapshot; classify the receipt as valid, invalid, or blocked; and update the graph only after Equation 2. Invalid runs remain immutable and visible. This design makes recovery operational: after a model, provider, or process interruption, the next action is derived from the last verified artifact rather than a reconstructed narrative.

3. Causal Study

3.1. Task and arms

We ran a frozen real-model study on a software-research audit derived from cross-dataset depression modeling. The task asks the agent to inspect a supplied evidence package and produce a report, a structured result, and a bounded reproducer. The source comprised 17 files (1,171,841 bytes). All registered arms used Codex CLI 0.144.0 with gpt-5.6-sol/ultra, the same public goal and hidden evaluator, seed 1101, isolated fresh workspaces, and frozen source bytes. The evaluator required a runnable report bundle and checked four metrics, a threshold count, and evidence paths.

The stock baseline (B0) used native goal execution with a 60k-token budget. C0 divided 60k nominal tokens across two candidate generators, a reviewer, and a repairer. After observing that neither produced a submission, we froze a follow-up C1 plan with 52k nominal stage budgets and added deliverable-first plus artifact gates. Finally, a 60k single-agent causal ablation retained only the deliverable-first instruction. Thus C1 is a mechanism-discovery follow-up, while the single arm is the sharper causal comparison to B0. Hidden scores were never used for generation or selection; all claims use the final frozen receipts.

Table 1. Frozen H-D2 results. “Correct” counts exact required numerical fields.

Arm	Tokens	Public	Artifact	Score	Correct
B0 stock	63,922	0/6	no	0	0/5
C0 portfolio	69,580	0/6	no	0	0/5
C1 gated	71,380	5/6	yes	65	0/5
D-first only	67,435	6/6	yes	65	0/5

3.2. Results

B0 investigated the task but produced no required file. C0 spent more compute yet both candidates, the review, and the repair also ended without a submission. In C1, the first scaffold created three runnable files after 10,817 tokens. That bundle survived to the end and received 65 points. The second scaffold, reviewer, and repair consumed a further 60,563 tokens but changed neither artifact nor score. The single deliverable-first arm independently produced a runnable package and the same 65 points without the portfolio.

The causal result is therefore not “more agents win.” The observed artifact-survival effect is localized to deliverable-first; the surrounding graph has measured marginal gain zero in this cell. This is useful positive knowledge: it identifies a smaller mechanism and falsifies an attractive, more expensive architecture.

The outcome also distinguishes three often-conflated objectives. Deliverability asks whether the required executable package exists; contract compliance asks whether its schema, reproducer, and evidence paths pass public checks; correctness is the independent hidden verdict. Deliverable-first improved the first two objectives, but not the third. A single scalar score would conceal this mechanism-level result.

AgentLoop’s evidence boundary also changed the scientific conclusion. Both 65-point artifacts used incorrect values for all four requested metrics and the threshold count. The single arm used 5.50% more tokens than B0, exceeding the preregistered 5% tolerance by 0.50 percentage points. We therefore retain deliverable-first as a mechanism hypothesis but do not claim universal superiority. Without fail-closed gates, a runnable public bundle and a nonzero score could easily have been reported as a win.

4. Discussion

Complexity must earn its place. Independent proposals are valuable only when they create information that a single run lacks. Reviewer and repair stages cannot improve an absent artifact, and they can consume the budget that would have completed one. AgentLoop makes topology a measured intervention rather than a

permanent commitment.

Auditability is a research capability. Separating engineering checks, development observations, and matched performance evidence makes negative results actionable. Immutable invalid trials expose concrete failure modes without being counted as candidate losses or rerolled until favorable. This supports cumulative improvement across model versions and process restarts.

Implications for autonomous research. Research agents must produce both an object of study and a defensible chain from intervention to claim. An early executable frontier reduces the risk that literature review, exploration, or debate uses the entire budget before measurement. Fail-closed receipts reduce the opposite risk: publishing a coherent artifact whose numbers are wrong. These controls are model-agnostic and can wrap a single agent or a changing graph; their value is testable without assuming that role count is a proxy for capability.

Limitations. The causal study covers one task family and one seed. B0 and the single ablation were not perfectly wrapper-matched, and the token difference narrowly exceeded tolerance. The 65-point artifacts were incomplete scientifically. Repeated wrapper-matched trials on a source-grounded retrieval engine and a deployable prediction engine, followed by a sealed unseen task, remain necessary. Our claim is about the demonstrated control mechanism and evaluation discipline, not broad performance dominance.

5. Conclusion

AgentLoop treats autonomous research as a contract-backed evidence process. In our study, an early runnable checkpoint—not a larger agent graph—caused deliverable survival, while external gates prevented that survival from being overstated as correctness. The broader lesson is simple: before delegating more roles, make the hypothesis falsifiable, preserve the first executable claim, and let independent evidence decide what remains in the loop.

References

- Jimenez, C. E., Yang, J., Wettig, A., Yao, S., Pei, K., Press, O., and Narasimhan, K. SWE-bench: Can language models resolve real-world GitHub issues? In International Conference on Learning Representations, 2024. URL <https://openreview.net/forum?id=VTF8yNQM66>.
- Schmidgall, S., Su, Y., Wang, Z., Sun, X., Wu, J., Yu, X., Liu, J., Liu, Z., and Barsoum, E. Agent laboratory: Using LLM agents as research assistants. arXiv preprint arXiv:2501.04227, 2025. URL <https://arxiv.org/abs/2501.04227>.
- Shinn, N., Cassano, F., Berman, E., Gopinath, A., Narasimhan, K., and Yao, S. Reflexion: Language agents with verbal reinforcement learning. In Advances in Neural Information Processing Systems, 2023. URL <https://arxiv.org/abs/2303.11366>.
- Tao, W., Zhou, Y., Wang, Y., Zhang, W., Zhang, H., and Cheng, Y. MAGIS: LLM-based multi-agent framework for GitHub issue resolution. In Advances in Neural Information Processing Systems, 2024. URL <https://openreview.net/forum?id=qevq3FZ63J>.
- Team Attention. Ralphthon at icml: Codex goal and auto research skill. Ralphthon participant presentation and guide, 2026. URL <https://ralphthon-icml-presentation.team-attention.com/slides.bi.html>. Accessed 2026-07-12.
- Wang, G., Xie, Y., Jiang, Y., Mandlekar, A., Xiao, C., Zhu, Y., Fan, L., and Anandkumar, A. Voyager: An open-ended embodied agent with large language models. Transactions on Machine Learning Research, 2023. URL <https://arxiv.org/abs/2305.16291>.
- Yang, J., Jimenez, C. E., Wettig, A., Lieret, K., Yao, S., Narasimhan, K., and Press, O. SWE-agent: Agent-computer interfaces enable automated software engineering. arXiv preprint arXiv:2405.15793, 2024. URL <https://arxiv.org/abs/2405.15793>.
- Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., and Cao, Y. ReAct: Synergizing reasoning and acting in language models. In International Conference on Learning Representations, 2023. URL <https://arxiv.org/abs/2210.03629>.

A. Frozen Goal Contract

The experiment used the following concrete instance of $G = (Y, E, C, I, S)$.

- Y : create a bounded report, structured result, and runnable reproducer for task hakbu-h-d2-immediate-v0.
- E : pass the six public artifact checks and an independent evaluator covering four metrics, the selected-threshold count, and evidence completeness.
- C : frozen source bytes, fixed model and reasoning effort, arm-specific nominal token budgets, no hidden-score selection, and a 5% B0-ablation compute tolerance.
- I : the 17-file evidence package, public goal, isolated workspace, Codex CLI 0.144.0, and gpt-5.6-sol/ultra.
- S : stop and retain the receipt on budget exhaustion, missing required artifact, evaluator failure, source or model mismatch, protected-answer exposure, or arm-specific infrastructure failure.

The frozen source aggregate, public goal, and hidden evaluator hashes were, respectively:

```
624235a091f0bb5fce02d2e523f8be8e3b6188d7be9ea9e58fef4e315f27798c
748cae1d282bac80fdca9a39449d9c86edfea4d130a19f6bf2b162e3653fb080
fe23eac41fd538a79cd2b8b66c220de50d5e1ae58ce21e92923c9cc543e3cd93
```

B. Full Execution Accounting

Table 2. Execution receipts. Wall time is end-to-end per arm; C0 and C1 sum stage wall times.

Arm	Nom.	Actual	$\Delta B0$	Wall (s)	Public/score
B0 stock	60,000	63,922	0.00%	161.465	0/6, 0
C0 portfolio	60,000	69,580	8.85%	203.654	0/6, 0
C1 gated	52,000	71,380	11.67%	216.911	5/6, 65
D-first only	60,000	67,435	5.50%	255.314	6/6, 65

C0 allocated nominal stage budgets of 18k, 18k, 8k, and 16k tokens. Both candidates ended at public verifier 0/6; no review file existed, and repair did not fabricate one. C1 allocated 10k, 10k, 6k, and 26k. Its first scaffold used 10,817 tokens and produced three runnable files at public 5/6. The remaining stages used 60,563 tokens; the controller retained the first scaffold because neither the artifact nor the score improved. The single arm used 67,435 tokens and reached public 6/6.

Both C1 and the single arm received external score 65. Their evaluator failure codes were evidence_incomplete, four metric_mismatch codes, and

threshold_count_wrong. Hence the observed mechanism is artifact survival and public contract compliance, not numerical correctness.

C. Receipt Integrity and Excluded Evidence

The plan hash was 2e82e0e449af905dbe3de2fbc034cfc39183b29735e89ad553abe7c44ac73851. The runner hashes were 3fa02413...23f5 (B0), 775ed5c0...9152 (C0), 2433c33b...0ab2 (C1), and 4d2c8562...abed (single). The complete 24-file manifest stores SHA-256 values for every receipt, ledger, transcript, and submitted file. Original evidence files were made read-only after freezing.

Five earlier attempts were excluded before interpreting performance: one stock run exceeded its compute polling boundary; two stock runs failed due to CLI/update packaging; one inherited an out-of-scope workspace instruction; and two C0 attempts stopped at zero tokens because of provider limits. Their receipts remain preserved, but none contributes a score to Table 1. This fail-closed policy prevents infrastructure outcomes from being relabeled as model or method outcomes.

D. Reproduction and Claim Boundary

Reproduction requires the frozen source package, goal, evaluator, runners, CLI executable, and model configuration named above. Run each arm in a fresh workspace, verify hashes before execution, keep generation blind to the hidden evaluator, and evaluate the final content-addressed artifact. Compare the resulting receipts without rerunning an unfavorable but valid outcome. The artifact accompanying this paper includes the exact evidence paths and a machine-readable hash manifest.

The claim ledger below separates observations from claims that remain blocked.

Table 3. Claim ledger for the reported cell.

Status	Claim
Supported	Deliverable-first caused a required runnable artifact to survive in H-D2 seed 1101.
Supported	The added C1 portfolio, reviewer, and repair produced zero artifact or score gain after the first scaffold.
Supported	Public contract compliance and hidden numerical correctness diverged in this cell.
Blocked	AgentLoop or deliverable-first universally outperforms stock native goal execution.
Blocked	The produced scientific measurements are correct.
Blocked	The comparison satisfies the preregistered B0-ablation 5% compute tolerance.

The next promotion-grade test is a wrapper-matched B0 versus deliverable-first comparison on fresh H-D2 seeds and a second task family, followed by a frozen candidate and sealed unseen task. Until those receipts satisfy Equation 2, deliverable-first remains a supported mechanism hypothesis rather than a universal winner.